

Information Processing Management

A Unified Privacy Framework for Secure and Utility-Preserving Retrieval-Augmented Generation

Rumaiza Kosar, Muhammad Imran, Muhammad Usman Yaseen
Department of Computer Science, COMSATS University Islamabad, Islamabad, Pakistan

Abstract

With the growth of the applications of Retrieval Augmented Generation (RAG) systems to extremely sensitive areas, including healthcare and finance, the privacy of Personally Identifiable Information (PII) and Protected Health Information (PHI) has become the center of more concerns. The RAG systems of the past retrieve unfiltered text and anonymize it after retrieval, meaning that sensitive data is still handled within the pipeline and can be leaked and extracted by adversarial users. This drawback led to the creation of a safer framework where privacy is ensured prior to any form of retrieval. LPRAG shows initial attempts at privacy preservation but is limited due to its use of document embeddings alone and making perturbations after the retrieval. As a result, LPRAG can minimize some of the leakage, although it does not essentially remove the risks existing in systems based on retrieval. The proposed methodology will fill these gaps by integrating privacy into the retrieval pipeline. Simultaneously, extracted entities and relations constitute a knowledge graph, which allows one to understand relations without having to store sensitive text in or retrieve it. Proposed system include hybrid retrieval, which involves using a perturbation of sensitive entities in conjunction with KG-based node and relation embeddings, such that the retriever can use structural context yet with a serious privacy guarantee. BLEU and ROUGEL scores are always higher in the case of hybrid retrieval than when using only the documents, which proves that the addition of KG information to document retrieval improves semantics and relevancy in the retrieval. Overall, the thesis proposes an effective and concise privacy-preserving variant of the RAG architecture that moves the anonymization to the first stage, employs the structured KG reasoning, and balances between privacy and utility with the help of the hybrid retrieval. The framework is more resistant to protection and has better retrieval performance than the standard methods, which makes it highly applicable to the sphere in which both confidentiality and the accuracy are required.

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1. Introduction

Retrieval-Augmented Generation (RAG) systems have significantly advanced the field of Natural Language Processing (NLP) by enabling large language models to dynamically retrieve external knowledge from databases, vector stores, and document repositories during response generation. This capability has greatly improved the contextual understanding, factual accuracy, and adaptability of generative AI systems across a wide range of applications. As a result, RAG-based architectures are increasingly being adopted in sensitive domains such as healthcare, finance, law, education, and enterprise environments, where real-time access to updated information is essential for decision-making and intelligent assistance [3, 5, 20]. Unlike traditional language models that rely solely on static training

data, RAG systems can continuously access external knowledge sources, making them more scalable and practical for real-world deployments.

Despite these advantages, the integration of external retrieval mechanisms introduces significant privacy and security concerns. Since RAG systems retrieve information directly from external repositories, there exists a considerable risk of exposing Personally Identifiable Information (PII), Protected Health Information (PHI), financial records, and other confidential data during both retrieval and generation stages [1, 6, 12]. Several recent studies have demonstrated that adversaries can exploit retrieval pipelines to extract sensitive information through carefully designed prompts, membership inference attacks, or knowledge extraction techniques [2, 14, 18, 26]. These vulnerabilities become even more critical in privacy-sensitive sectors where data confidentiality and regulatory compliance are mandatory requirements. In healthcare systems, unauthorized exposure of patient records may violate legal regulations and ethical standards, while in finance, leaked transactional information can result in severe security and reputational consequences.

Traditional RAG architectures primarily focus on improving retrieval relevance and generation quality but often neglect privacy-preserving mechanisms during the retrieval stage itself. Most existing approaches apply anonymization or perturbation after sensitive information has already been retrieved, which leaves a critical privacy gap within the retrieval pipeline [1, 17]. Consequently, confidential data may still be exposed internally before masking or sanitization occurs. Recent research has further highlighted multiple attack vectors against RAG systems, including retrieval database leakage, prompt injection, knowledge poisoning, and inference-based extraction attacks [7, 13, 16]. These findings indicate that privacy risks in RAG systems are not limited to output generation alone but are deeply connected to how knowledge is retrieved, stored, and processed.

To address these challenges, researchers have proposed several privacy-preserving RAG frameworks, including differential privacy-based retrieval, secure embedding generation, federated retrieval mechanisms, and synthetic data approaches [10, 12, 17, 21, 24]. Although these methods improve privacy protection to some extent, they often introduce a difficult privacy-utility trade-off. Stronger privacy constraints can reduce retrieval precision and negatively affect the quality of generated responses, while highly accurate retrieval may increase the risk of sensitive data exposure. This trade-off remains one of the most significant open challenges in privacy-aware RAG research, particularly for applications where high accuracy and reliability are critical.

The proposed framework in this research addresses these limitations by integrating privacy protection directly into the retrieval pipeline rather than relying solely on post-retrieval sanitization techniques. The framework combines entity recognition, sensitive attribute detection, and knowledge graph (KG) construction to ensure that confidential information is identified and protected before retrieval occurs. Instead of storing raw sensitive content within retrievable documents, the system transforms sensitive entities into privacy-aware representations, thereby reducing the possibility of direct information leakage. This proactive approach significantly strengthens privacy preservation while maintaining retrieval effectiveness.

A major contribution of the proposed framework is the integration of a hybrid retrieval mechanism that combines document-based semantic embeddings with knowledge graph-based retrieval. Knowledge graphs provide structured representations of entities and their relationships, enabling semantic reasoning, contextual understanding, and controlled access to external knowledge [22, 25]. Unlike conventional document-only retrieval systems, the hybrid architecture allows the model to retrieve contextually relevant information through both semantic similarity and graph-based relational reasoning. This not only improves retrieval accuracy but also reduces unnecessary exposure to raw textual data. Furthermore, knowledge graph embeddings enhance the system’s ability to preserve contextual integrity while maintaining privacy-aware retrieval operations.

In addition to structured retrieval, the framework incorporates multiple privacy-preserving techniques, including masking, attribute coarsening, and differential privacy mechanisms. Masking replaces sensitive values with generalized placeholders, while attribute coarsening reduces the specificity of sensitive attributes to minimize re-identification risks. Differential privacy introduces controlled statistical noise into embeddings and retrieval representations, ensuring that individual records cannot be easily inferred from system outputs [12, 17]. Together, these methods form a Unified Privacy Policy (UPP) that balances privacy protection with retrieval utility. The integration of these mechanisms before the retrieval stage provides stronger protection compared to traditional approaches that apply privacy controls only after retrieval.

Another important aspect of this framework is its resilience against emerging RAG-specific attacks. Recent studies have shown that adversaries can exploit retrieval databases through membership inference attacks, malicious retrieval

poisoning, and prompt-based leakage strategies [13, 14, 18]. The proposed approach mitigates these risks by enforcing controlled retrieval pathways and restricting direct exposure to sensitive data representations. Additionally, the use of structured knowledge graphs reduces dependence on unrestricted document retrieval, thereby limiting opportunities for adversarial exploitation. This makes the framework more robust and suitable for deployment in real-world environments where security and privacy are essential.

The effectiveness of the proposed framework is evaluated using both privacy-oriented and utility-oriented metrics. Privacy performance is assessed through residual leakage analysis, exposure probability, and resistance against inference attacks, while utility is measured using standard evaluation metrics such as BLEU and ROUGE-L scores. Experimental findings demonstrate that the proposed privacy-preserving hybrid RAG framework achieves stronger privacy guarantees while maintaining high response quality and contextual relevance. Comparative analysis with traditional RAG systems further highlights the framework’s ability to significantly reduce sensitive information exposure without causing substantial degradation in model performance.

Overall, this research contributes to the growing field of secure and privacy-aware Retrieval-Augmented Generation by proposing a structured hybrid framework that integrates privacy protection, semantic retrieval, and knowledge graph reasoning within a unified architecture. The framework not only addresses critical privacy challenges associated with modern RAG systems but also enhances retrieval reliability, contextual understanding, and robustness against adversarial attacks. These characteristics make the proposed approach highly suitable for high-stakes domains such as healthcare, finance, legal systems, and enterprise AI applications, where preserving both privacy and performance is equally important.

2. Related Work

The field of privacy-preserving techniques in Retrieval-Augmented Generation (RAG) systems has evolved rapidly in recent years, driven by the increasing adoption of large language models in privacy-sensitive domains such as healthcare, finance, legal services, government systems, and enterprise intelligence platforms. Researchers have explored multiple approaches to safeguard sensitive information while maintaining the effectiveness and contextual relevance of generated responses. Existing literature highlights that although RAG architectures significantly enhance generative capabilities through external knowledge retrieval, they simultaneously introduce new privacy, security, and data governance challenges [3, 5, 7, 20]. Consequently, privacy-preserving RAG has emerged as a critical research direction aimed at balancing retrieval utility with strong confidentiality guarantees and regulatory compliance.

Early privacy-preserving frameworks primarily focused on synthetic data generation and anonymization techniques to reduce the exposure of sensitive information. Approaches such as the SAGE framework introduced synthetic replacement strategies in which sensitive attributes and semantic relationships are transformed into privacy-preserving synthetic representations while preserving the structural integrity of the original data. These frameworks generally operate in multiple stages, beginning with semantic extraction and attribute synthesis, followed by refinement procedures that use iterative feedback mechanisms to improve consistency and utility. Synthetic data generation has shown promising results in reducing direct information leakage; however, maintaining semantic fidelity and contextual coherence across generated samples remains a significant challenge [24]. In many cases, excessive perturbation or synthetic replacement negatively impacts retrieval precision and weakens the contextual quality of generated responses. Recent studies therefore suggest integrating entity-aware contextualization and knowledge graph reasoning to preserve semantic relationships while enforcing privacy constraints [1, 8].

Differential Privacy (DP) has become one of the most widely accepted and theoretically grounded privacy-preserving mechanisms in machine learning and RAG systems. Differential privacy protects individual data records by introducing controlled statistical noise, ensuring that the presence or absence of a particular data sample does not significantly influence model outputs [12, 17]. Several recent studies have adapted DP mechanisms specifically for retrieval-augmented architectures. For example, Privacy-Aware Decoding (PAD) introduces Gaussian noise during token decoding to minimize the probability of exposing sensitive information in generated outputs [11]. PAD is particularly beneficial because it does not require complete retraining of the underlying language model, making it computationally efficient and scalable for large deployments. Similarly, token-level differential privacy approaches allocate privacy budgets to sensitive tokens or entities, enabling fine-grained control over privacy protection. Although these techniques improve confidentiality, they often introduce degradation in fluency, coherence, and semantic relevance, further emphasizing the persistent privacy-utility trade-off within RAG systems.

Recent studies have also investigated local differential privacy (LDP) techniques for embedding-level protection in retrieval systems. Unlike traditional differential privacy approaches that apply perturbation after data aggregation, LDP mechanisms add privacy-preserving noise directly to embeddings before they are stored or retrieved, thereby minimizing exposure risks during retrieval operations [1, 10]. Embedding perturbation strategies are particularly important because vector databases form the backbone of modern RAG pipelines. However, research indicates that poorly protected embeddings remain vulnerable to inversion attacks, reconstruction attacks, and semantic leakage [14, 16, 18]. Consequently, several researchers advocate embedding-aware privacy protection that preserves semantic similarity while preventing sensitive data reconstruction.

Beyond differential privacy, researchers have explored secure embedding techniques, perturbation-based retrieval mechanisms, and retrieval isolation architectures. Methods such as Private Retrieval-Augmented Generation with Random Projection and privacy-preserving federated embedding learning attempt to obscure sensitive semantic representations before retrieval occurs [10, 21]. These techniques aim to preserve contextual retrieval quality while preventing adversaries from reconstructing sensitive information from embedding spaces. Federated retrieval systems further decentralize data storage and retrieval, allowing multiple parties to collaboratively train or retrieve information without directly exposing raw datasets [21]. Although these methods strengthen confidentiality, they often introduce synchronization overhead, increased computational complexity, and scalability limitations in real-world enterprise deployments.

In parallel, adaptive privacy and security enforcement mechanisms have gained attention as practical solutions for enterprise-scale RAG deployments. Recent frameworks propose dynamically adjusting privacy protections according to contextual risk, user authorization level, and query sensitivity [6, 23]. Adaptive RAG systems incorporate contextual risk analysis, role-based access control, dynamic auditing, and compliance-aware retrieval mechanisms to ensure alignment with regulatory standards such as GDPR, HIPAA, and enterprise governance policies. These systems attempt to provide flexible privacy enforcement without uniformly restricting retrieval quality. Enterprise-oriented architectures are increasingly moving toward isolated and on-premise deployment environments, where sensitive organizational data remains under local control rather than being exposed through public cloud infrastructures.

Another significant advancement in the literature involves encrypted and distributed retrieval frameworks. Systems such as RemoteRAG integrate Partial Homomorphic Encryption (PHE) with differential privacy techniques to secure query transmission and embedding retrieval while preserving retrieval efficiency [19]. Similarly, distributed graph-based retrieval systems and edge-cloud retrieval architectures attempt to decentralize sensitive information storage, reducing the risk of centralized data leakage [22]. Distributed retrieval frameworks also improve system robustness and scalability by reducing reliance on single-point retrieval infrastructures. However, encryption-based approaches may still suffer from computational overhead and latency issues when handling large-scale retrieval operations.

The growing adoption of knowledge graphs in RAG systems has contributed significantly to privacy-aware retrieval research. Knowledge graphs provide structured representations of entities and their relationships, enabling semantic reasoning, contextual understanding, and controlled knowledge access [22, 25]. Compared to conventional document-only retrieval, graph-based retrieval reduces dependency on raw textual content and minimizes uncontrolled exposure of sensitive information. Studies on graph retrieval-augmented generation have demonstrated that knowledge graphs improve retrieval explainability and contextual precision while enabling stronger privacy controls through entity-level management [9]. Furthermore, hybrid retrieval architectures that combine vector embeddings with graph reasoning have shown improvements in BLEU and ROUGE-L performance metrics, particularly in complex question-answering tasks where contextual reasoning is essential.

Recent literature has also emphasized the importance of secure and isolated retrieval pipelines for enterprise AI systems. Privacy-Aware RAG frameworks propose secure retrieval isolation, restricted retrieval access, and compartmentalized knowledge storage to prevent unauthorized data exposure [23]. Similarly, provably secure RAG architectures attempt to establish formal security guarantees by designing mathematically verifiable retrieval protocols [28]. These approaches aim to ensure that sensitive knowledge cannot be reconstructed or inferred even if adversaries gain partial access to the retrieval infrastructure. Although such frameworks improve theoretical security guarantees, they often require specialized architectures and increased computational resources, limiting their practical applicability in resource-constrained environments.

In addition to privacy concerns, adversarial threats against RAG systems have become a major focus of recent research. Studies such as RAG-Thief have demonstrated that attackers can exploit retrieval pipelines through carefully crafted agent-based attacks to extract sensitive knowledge from external databases [2]. Membership inference attacks

further reveal that adversaries can determine whether specific data samples exist within retrieval databases by analyzing model outputs and retrieval behaviors [14, 18, 26]. Other studies have investigated knowledge poisoning attacks, where malicious or manipulated retrieval content is injected into external knowledge sources to influence generated responses [13]. Multi-modal RAG systems have also been shown to introduce additional vulnerabilities by exposing sensitive information through images, metadata, and cross-modal representations [27]. These findings collectively highlight that RAG security must be approached as a multi-layered problem involving retrieval security, embedding protection, generation safety, adversarial robustness, and access control simultaneously.

Another emerging concern identified in recent literature is the issue of privacy leakage through semantic associations and contextual reasoning. Even when direct identifiers are removed, modern language models may infer sensitive attributes through correlated contextual information and relational dependencies [3, 9]. This challenge becomes particularly significant in healthcare and legal applications, where seemingly harmless contextual cues may reveal confidential information indirectly. Researchers therefore emphasize the need for context-aware privacy enforcement mechanisms capable of identifying hidden semantic relationships that could lead to unintended disclosure.

Furthermore, recent research has identified what is commonly referred to as the “RAG paradox,” where access to external knowledge significantly improves model accuracy and reasoning capabilities while simultaneously increasing attack surfaces and privacy risks. The retrieval component that enhances contextual understanding also becomes the primary channel through which sensitive information leakage can occur. Consequently, researchers increasingly advocate for retrieval-aware privacy enforcement mechanisms that proactively protect data before retrieval rather than relying solely on output-level sanitization [1, 11, 17]. This shift from reactive privacy protection toward proactive privacy-preserving retrieval forms the conceptual foundation of many modern secure RAG architectures.

Despite substantial progress, the literature reveals that existing privacy-preserving RAG frameworks remain fragmented and domain-specific. Most proposed systems focus on isolated techniques such as differential privacy, encryption, synthetic data generation, federated retrieval, or secure embeddings independently, without integrating them into a unified architecture. As a result, many solutions struggle to generalize across heterogeneous domains and varying privacy requirements. Systems designed for healthcare applications may not scale efficiently to finance or legal environments, while enterprise-focused solutions often prioritize strict security at the cost of retrieval flexibility and model performance [6, 19, 23]. Moreover, current methods rarely provide unified privacy governance capable of adapting dynamically to different sensitivity levels, user roles, and contextual requirements.

Overall, the existing body of literature demonstrates considerable progress in privacy-preserving retrieval-augmented generation through techniques such as differential privacy, synthetic data generation, encrypted retrieval, federated learning, adaptive access control, secure embeddings, and knowledge graph reasoning. However, most existing approaches provide only partial solutions and fail to achieve an optimal balance between privacy, scalability, contextual relevance, retrieval efficiency, and adversarial robustness. The absence of a unified privacy policy capable of operating consistently across heterogeneous domains remains a critical limitation in current research. To address these gaps, this research proposes a comprehensive privacy-preserving RAG framework that integrates hybrid retrieval, knowledge graph reasoning, entity-aware privacy enforcement, adaptive privacy policies, and multi-layered security mechanisms within a unified architecture. The proposed framework aims to provide stronger privacy guarantees while maintaining high retrieval utility, contextual accuracy, scalability, and adaptability across diverse application domains such as healthcare, finance, legal systems, and enterprise AI platforms. ““

3. Methodology

This research aims to design a robust, privacy-preserving information retrieval system tailored for sensitive textual data, specifically in domains such as medical and social-health records. The objective is to provide a systematic approach for efficiently extracting, organizing, and retrieving sensitive information while ensuring stringent privacy protection through advanced anonymization techniques. The methodology integrates cutting-edge Natural Language Processing (NLP) models, Knowledge Graph (KG) construction, differential privacy mechanisms, and hybrid retrieval strategies to achieve a balance between utility and privacy.

3.1. Overview of the Proposed System Model:

The system follows a modular and multi-stage design, where each stage processes raw textual data systematically to convert it into structured knowledge representations, while ensuring privacy preservation throughout the process.

The proposed methodology effectively handles two critical challenges:

1. Extracting valuable information from unstructured, dense text.
2. Ensuring the privacy of personally identifiable information (PII) and sensitive content.

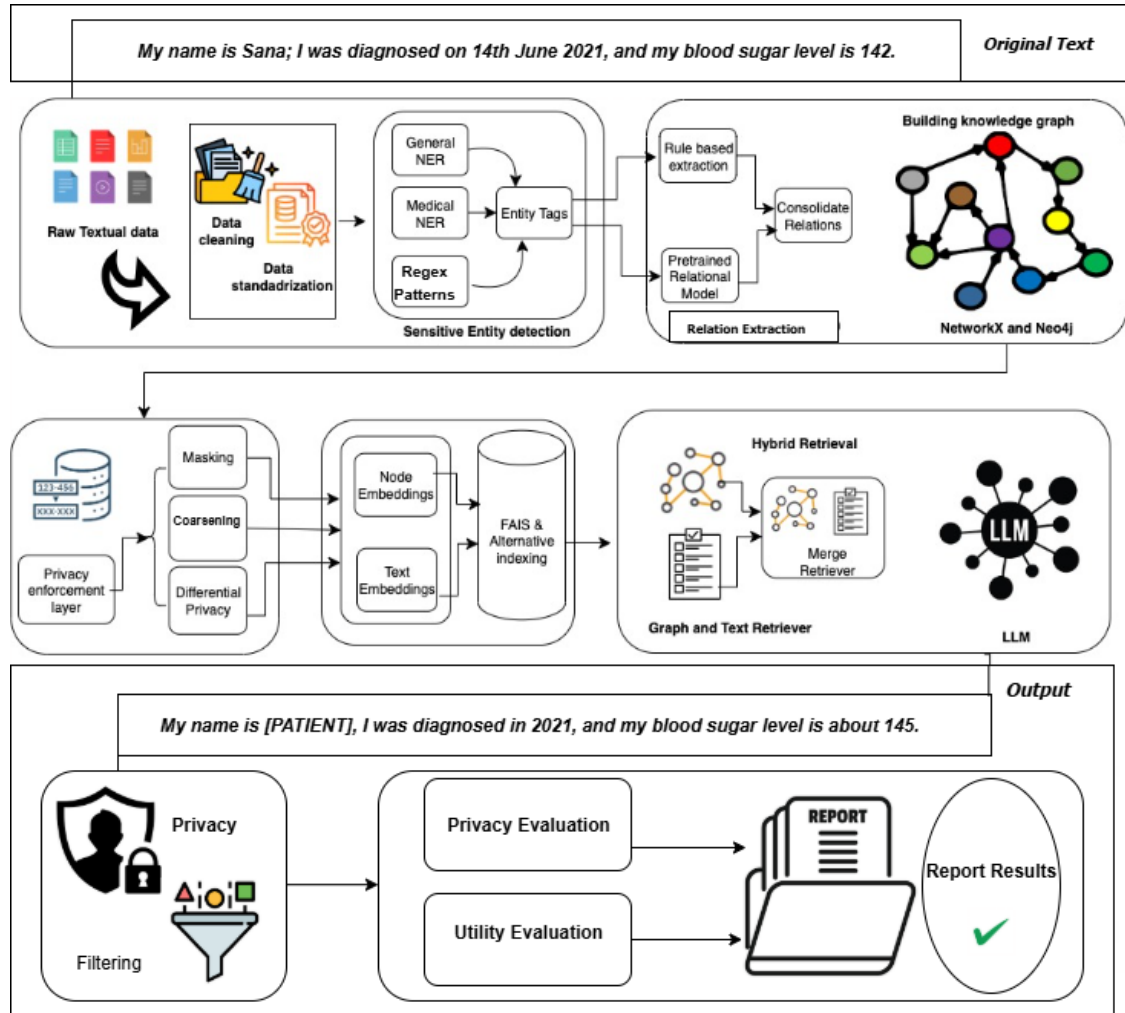


Figure 1. Methodology Diagram

The proposed methodology follows these sequential steps:

- **Data Preprocessing and Normalization:** The input raw text is cleaned, normalized, and standardized to ensure consistency, accuracy, and quality across the dataset.
- **Information Extraction:** Named Entity Recognition (NER) and Relation Extraction (RE) techniques are applied to categorize entities and identify semantic relations between them. This step transforms raw text into a structured knowledge representation.
- **Knowledge Graph Construction:** A semantic Knowledge Graph (KG) is built to represent entities and their relationships. The nodes in the graph correspond to entities, and the edges represent the relationships, enabling relational and structural reasoning.

- **Utility-Preserving Privacy Application:** Privacy-preserving techniques, such as entity masking, coarsening, and differential privacy, are applied to sensitive entities. These methods ensure that privacy is maintained while preserving the data’s utility for retrieval.
- **Embedding and Indexing:** The documents and KG nodes are converted into dense semantic vectors using pre-trained models. These embeddings are indexed for fast retrieval using algorithms like FAISS (Facebook AI Similarity Search).
- **Hybrid Retrieval:** A hybrid retrieval mechanism is employed, combining document-level and KG-level retrieval for more context-sensitive and semantically rich results.
- **Utility Analysis:** Utility metrics, such as BLEU and ROUGE-L, are used to evaluate the trade-off between privacy and the usefulness of the retrieved data.

[H] Privacy-Preserving Knowledge Graph Construction and Hybrid Retrieval

Input:

- D_{raw} : A collection of raw textual documents (e.g., medical records).
- Q_{raw} : A user query text.
- k : Number of top results to retrieve.
- ϵ : Privacy budget for Differential Privacy (Laplace noise).
- α : Weighting parameter for hybrid retrieval (balance between document and KG node scores).
- NER_Model: Pre-trained Transformer model for Named Entity Recognition.
- RE_Model: Pre-trained Transformer model for Relation Extraction (with fallback heuristics).
- SentenceTransformer: Pre-trained model for generating embeddings.
- WordNet, spaCy: Libraries for symbolic knowledge (if used in RE fallback).

Output:

- Results: A ranked list of k privacy-preserved documents and/or KG nodes relevant to Q_{raw} .
- UtilityMetrics: BLEU and ROUGE-L scores evaluating UPP impact.

3.2. Dataset Description:

The framework was tested using multiple datasets to assess its performance in privacy-sensitive retrieval and question answering tasks. The following datasets were selected for testing the framework:

Table 1. Summary of Datasets

Dataset	Domain	Size	Format
HealthCareMagic (HCM)	Medical QA	~200,000 Dr-PT conversations	Dialogues
Natural Questions (NQ)	Open-domain QA	~300,000 questions	Question + passage pairs
HotpotQA	Open-domain QA	~113,000 QA pairs	Multi-hop QA with passages
SQuAD	Open-domain QA	~100,000 QA pairs	Paragraphs with span-based QA

3.2.1. *HealthCareMagic (HCM):*

HealthCareMagic (HCM) is a massive medical question and answer collection of over 200,000 doctor-patient conversations. This data has a broad spectrum of issues concerning healthcare, and as such, it would be a valuable resource in measuring how well the framework would operate in regards to privacy and delivery of valuable answers to sensitive questions in health-related matters.

- **Content Overview:** The HCM discussions are not general (e.g. symptoms, treatment, medication consultation) only but also medical (e.g. post-surgery instructions, diagnosis suggestions). Medical counsel may be extremely confidential, including health status of patients, medical history, drugs used, health issues and many more.
- **Privacy Challenge:** This data is significant in the testing how the framework could safeguard sensitive information relating to health matters during the retrieval process. Medical information is extremely sensitive and, therefore, the privacy laws (e.g. HIPAA) provide its protection. One of the most important aspects is to consider the possibility of the framework to avoid privacy leakage and be accurate in question-answering (QA) tasks.
- **Use Case:** HCM is the main testbed of evaluating the efficiency of the framework in terms of privacy protection versus retrieval accuracy. It is used to assess the privacy-preserving techniques in addressing the potential leakage of sensitive health data, and yet obtaining the most relevant and accurate responses to medical queries.

3.2.2. *Natural Questions (NQ) + Snippets HCM:*

The Natural Questions (NQ) data set is comprised of approximately 300,000 question-answer pairs, which include the wide range of general knowledge and the open domain questions. HCM snippets (sensitive medical information) were inoculated into the dataset in order to evaluate the privacy preservation retrieval capabilities of the framework.

- **Content Overview:** NQ is founded on real life questions whose answers are not always readily accessible using straightforward lookups. The code injected HCM snippets are complicated since they contain personal health information, including doctor patient dialogue, prescriptions, and other sensitive health information.
- **Privacy Challenge:** The main issue with this data is that the information is of both open and personal nature of health-related data. The framework should be able to identify the difference between publicly available information and sensitive information so that the sensitive information does not accidentally find its way into the answer to a query.
- **Use Case:** The dataset is applied to test fact based QA where sensitive information is mixed with general information. The framework should ensure that sensitive data (e.g. patient data) are safe during retrieval and at the same time give the right and the correct answers. This data is also a challenge to the system in that the system must not violate privacy and at the same time offer high quality information retrieval.

3.2.3. *HotpotQA + HCM Snippets:*

HotpotQA dataset is an open-domain QA dataset that consists of approximately 113,000 question-answer pairs. It works on multi-hop questions, in which the model is required to answer a question using multiple passages. HCM snippets were injected in this experiment to add sensitive medical information into the dataset.

- **Content Overview:**
HotpotQA problems are multi-hop reasoning tasks, which are characterized as complex, and in this task, the model has to read multiple passages to produce an answer. The dataset was also made more relevant to privacy-preserving assessments by adding sensitive medical snippets of HealthCareMagic.
- **Privacy Challenge:** This is because the questions are multi-hop. Since this model uses various sources of information, it should not reveal sensitive information in the HCM snippets in the process. The privacy should be ensured at every step of retrieving information and this increases the complexity of the task.
- **Use Case:** The HotpotQA testing framework uses HCM snippets to test the multi-hop QA feature of the framework. The model is requested to search and combine information on a variety of documents, and privacy should not be lost at any stage of the process. The problem is to use the privacy-preserving methods successfully even in the cases when information is supplied by multiple sources.

3.2.4. *SQuAD + HCM Snippets:*

Stanford Question Answering Data set (SQuAD) is a QA dataset made of approximately 100,000 pairs of questions and answers, comprising of a comprehension-based dataset. This dataset was adapted to test the speed and effectiveness of the model in comprehension based tasks with sensitive data with the introduction of the HCM snippets.

- **Content Overview:**

SQuAD targets the span-based QA, in which the model should recognize specific textual spans in order to answer questions. To test the privacy-preserving QA properties of the framework, the dataset was augmented with the snippets of HCMs, containing sensitive medical data.

- **Privacy Challenge:**

The main challenge is to safeguard sensitive health information while enabling the model to comprehend and respond to questions based on the relevant text spans. Since the task is span-based, sensitive data must be retained within the selected span used in the answer.

- **Use Case:** This dataset presents a problem to the understanding capabilities of the framework concerning sensitive health data. It evaluates the ability of the system to respond to comprehension based questions but maintain privacy to make sure that sensitive information is not disclosed in the response.

3.3. *Data Preprocessing and Standardization:*

Raw text data undergoes a series of cleaning and standardization steps to ensure consistency and prepare it for subsequent NLP tasks. The preprocessing pipeline includes the following steps:

- **Unicode Normalization:** Every text is transformed into a canonical form with the NFKC (Normalization Form KC) scheme, which provides that characters that visually appear the same but are different in their encoding are treated the same way.
- **Whitespace Regularization:** Consecutive whitespaces are also changed to one space and any leading space or trailing space are also removed to ensure uniformity in tokenizing.
- **HTML Tag Removal:** Regular expressions are used to eliminate any embedded HTML or XML tags that might be present in the raw data and only meaningful text is retained.
- **Punctuation Stripping:** Redundant punctuation marks, especially the ones that come at the end of the sentence, are eliminated in order to enhance linguistic readability and prevent noisyness during token level analysis.

3.4. *Information Extraction:*

After cleaning up the text, it then applies sophisticated NLP to get organized information. This will provide the ground work of construction of the knowledge graph that is, fundamentally converting raw sentences into understandable and significant fragments which we can actually utilize.

- **Named Entity Recognition (NER):** It uses a pre-trained Hugging Face Transformer model for entity recognition. The most commonly used is "d4data/biomedical-ner-all" because it is ideal for picking out biomedical entities, such as diseases, chemicals, medical procedures, parts of the body, among many others. If that model isn't available for some reason, the default falls to the general-purpose model "dslim/bert-base-NER", which works quite well, too. The NER pipeline takes advantage of an extremely simple aggregation strategy to combine multi-word entities into neat, stand-alone units. For every detected entity, it captures the text, category label—that is, DISEASE, CHEMICAL, or PERSON and character start and end positions. The result will be a clean, well-structured list of the most important concepts in the text.

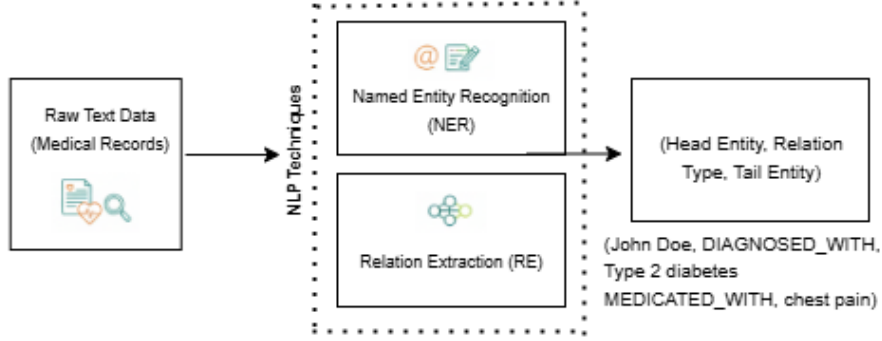


Figure 2. Information Extraction Process: NER and RE

- **Relation Extraction (RE):** Once identified, the next step taken by the system is to get down to the relations among these entities. It attempts to utilize a pre-trained Transformer sequence classification model: "mrmm8488/bert-mini-finetuned-relation-extraction". This model takes as input sentences in which two candidate entities are marked with special tokens for example, [E1] and [E2] and predicts what kind of relation exists there between them, such as MEDICATED WITH, VISITED, or DIAGNOSED WITH. To put it in simple terms, it is like this: Given this sentence and these two underlined entities, what is their relation? If this fancy model is not available, or should it fail to find a relation, the system simply doesn't give up. It, instead, invokes a fallback approach by keyword matching, looking for words such as "prescribed", "visited", or "diagnosed" in the text, and matches those with the type of entities in order to guess the relationship. All these relations reduce to simple semantic triplets in the form of (Head Entity, Relation Type, Tail Entity). A typical example could be (John Doe, DIAGNOSED WITH, Type 2 diabetes) or (ibuprofen 200 mg, MEDICATED WITH, chest pain). These constitute the nodes and edges of the knowledge graph that follows.

3.5. Knowledge Graph Construction:

The Knowledge Graph (KG) captures the relationships between entities, represented as nodes and edges. The graph allows for efficient relational reasoning over the data, supporting improved information retrieval.

$$KG = (E, R) \quad (1)$$

Where:

- $E = \{e_1, e_2, \dots, e_n\}$ is the set of entities.
- $R = \{(e_i, r_{jk}, e_k) \mid e_i, e_k \in E, r_{jk} \in RelationTypes\}$ is the set of relations between entities.

Working with graphs in Python gets a whole lot easier once you start using networkx. What's really going on is that any time you pull out an entity during information extraction, that thing becomes a node in your graph. Each node gets an ID built from the document's ID, the entity's text, and its semantic label. So you always know exactly where each node came from. On top of that, every node holds onto a few essentials its text, label, and doc_id|soyouneverlosesightofthebasics.

3.6. Utility-Preserving Privacy (UPP) Framework

Accordingly, a formal framework of UPP has been systematically applied to the original text and the cleaned text in order to deal with the sensitive requirement of protecting patient information. This framework was designed with great care to anonymize the data by integrating many techniques that enhance privacy, such as entity masking and data coarsening, with the stringent application of differential privacy. It will attempt to ensure a balance in protection and retain enough utility of the data for further analysis and retrieval. The composite privacy transformation function $D_{perturbed} = UPP(D_{raw})$ is the central operation of this module.

3.6.1. Entity Masking and Coarsening

This is the first layer of protection for privacy and conducts a systematic masking or coarsening of sensitive entities previously detected through the NER module. It does so by iterating over the identified entities in reverse order of occurrence within the text. In such a way, processing entities in reverse order avoids shifts of textual offsets that might invalidate positions of entities subsequently processed.

- **Categorical Masking:** This involves the replacement of direct identifiable information and sensitive categories with placeholder tokens. Examples are the systematic replacement of names of PERSON, names of ORGANIZATIONS, precise LOCATION details, and broadly classified DISEASE labels with the generalized tokens [PERSON], [ORG], [LOCATION], and [DISEASE] respectively. In this way, specific instances are redacted while retaining the semantic category.
- **Date Coarsening:** The fine-grained date information, such as specific YYYY-MM-DD formats, gets generalized to retain only the year, thus becoming [DATE:YYYY], or gets reduced to the generic token [DATE] when the year context is not sufficient. Additional regular expression patterns are run to find and coarsen other explicit date expressions that might have slipped through NER.
- **Numeric ID Coarsening:** Long sequences of digits, usually standing for patient IDs, record numbers, or other sensitive numerical sequences, are replaced by the non-specific [NUM] token. This prevents direct linkage to individuals through arbitrary identification numbers.

The first masking and coarsening step may be given the following mathematical expression:

$$D_{masked} = \text{MaskAndCoarsen}(D_{raw}, \text{Entities}, \text{RegexRules}) \quad (2)$$

Where D_{raw} is the original text, Entities are the identified NER spans, and RegexRules are the patterns for date and numeric ID coarsening.

3.6.2. Differential Privacy for Numeric Values

Application of differential privacy to residual numerical values in masked text will guarantee mathematically provable layers of privacy. The mechanism works by perturbing numeric data in a manner such that the presence or absence of any single individual's data in the dataset will not significantly change the output and thus protect individual privacy.

- **Integer Perturbation:** All the integers identified in the already masked and coarsened text undergo perturbation. This happens through the addition of random noise, which is drawn from a Laplace distribution. The Laplace mechanism is chosen because it can be applied directly in adding noise to numerical outputs for the satisfaction of ϵ -differential privacy.
- **Privacy Budget (ϵ):** This is the key parameter that controls the scale of noise added, quantifying the privacy budget. The smaller the value of ϵ , the larger the scale of noise enforced, hence providing stronger privacy guarantees. On the other hand, larger ϵ yields less noise, hence weaker privacy but generally higher utility. This is the direct control over the privacy/utility trade-off.
- **Sensitivity (Δf):** A *sensitivity* parameter (typically set to 1.0 for count queries or single-value perturbation) is used to scale the Laplace noise according to the maximum possible change an individual record can introduce to the numeric value. The scale of the Laplace distribution, b , is calculated as $b = \Delta f / \epsilon$.

The addition of Laplace noise to a numeric value v can be formalized as:

$$v_{perturbed} = v + \text{Laplace}\left(\frac{\Delta f}{\epsilon}\right) \quad (3)$$

Where $\text{Laplace}\left(\frac{\Delta f}{\epsilon}\right)$ denotes a random variable drawn from a Laplace distribution with scale parameter $\frac{\Delta f}{\epsilon}$.

This multi-step process of privacy transformation is orchestrated in the system by the function `apply_ai_uvp`. It triggers the masking and coarsening processes before applying the Laplace noise perturbation to all identified numeric tokens. The texts that come out of this comprehensive UPP framework, henceforth called perturbed texts, are used for all subsequent embedding and retrieval processes. In this way, all information processed downstream is guaranteed to meet the prescribed privacy standards.

[H] Utility-Preserving Privacy (UPP) Framework

Input:

- $D_{cleaned}$: A single cleaned textual document.
- `NER_Spans`: A list of named entities detected in $D_{cleaned}$, with their text, label, and offsets.
- ϵ : Privacy budget for Differential Privacy (Laplace noise).

Output:

- $D_{perturbed}$: The privacy-preserved (perturbed) version of the document.

3.7. *Embedding and Indexing*

Solutions will involve mapping both the privacy-preserved documents and nodes of the knowledge graph to dense, high-dimensional vector representations, or embeddings, enabling effective and semantic similarity search, then arranging and storing them in special indexes devised for speedy retrieval.

3.7.1. *Sentence Embeddings*

Embeddings of high quality represent the semantic meaning of documents of text and single members of the knowledge graph. To this end, the pre-trained model of a Sentence Transformer, the label of which is all-MiniLM-L6-v2, will be used. It is favored due to its efficiency, as well as its ability to produce semantically rich representations of vectors, therefore, making it possible to use it to perform effective tasks of semantic similarity.

- **Document Embeddings:** All these perturbed versions of the documents are inputs of the UPP framework that are sent through the Sentence Transformer. It generates a predetermined-size vector of each document indicating the combined surrounding meaning. These vectors consist as a set of the *doc_embs* matrix, and each row of which is the embedding of a document. This could be represented as:

$$E_{doc_i} = \text{SentenceTransformer}(D_{perturbed_i}) \quad (4)$$

Where $D_{perturbed_i}$ is the i -th privacy-preserved document, and E_{doc_i} is its corresponding embedding vector.

- **Knowledge Graph Node Embeddings:** On the same note, the textual descriptions of each node in the Knowledge Graph (node texts) are also converted into embeddings (node texts). This operation makes sure that the meaning of the separate entities, semantically (e.g., diabetes type 2, Mayo Clinic), is represented in a document embedding-compatible [vector space]. These constitute the matrix of the node embs, i.e. the *node_embsmatrix*.

$$E_{node_j} = \text{SentenceTransformer}(NodeText_j) \quad (5)$$

Where $NodeText_j$ is the text of the j -th KG node, and E_{node_j} is its embedding vector.

All generated embeddings, be it *doc_embs* or *node_embs*, are L2-normalized. This is an important step because it rescales each vector to unit magnitude; such standardization of magnitudes ensures that any subsequent similarity calculations, such as cosine similarity, depend exclusively on the angular difference between vectors and not on their raw magnitudes. This consistency is important for proper similarity comparisons.

3.7.2. Index Creation

To enable fast and scalable nearest neighbor searches, a separate index is created for each embedding set. Indexing technology choice depends on availability and performance requirements:

- **FAISS Indexing:** If the `faiss` library is available, highly optimized - `faiss.IndexFlatIP` (Inner Product) indexes are used. The FAISS library is considered the most efficient solution for doing similarity searches on big datasets of dense vectors, offering great speed-ups compared to brute-force comparisons. In FAISS, inner product similarity of L2-normalized vectors directly corresponds to cosine similarity.
- **Scikit-learn Fallback:** In environments where `faiss` is not installed or available, `sklearn.neighbors.Nearest - Neighbors` is employed as a robust fallback mechanism. Usually considerably slower for very large datasets compared to FAISS but gives a reliable way of doing approximate nearest neighbor searches based on cosine distance—which is readily convertible to cosine similarity by $1 - distance$.

The creation of such specialized indexes enables the queries' effective matching against the large collection of privacy-preserved documents and knowledge graph entities, which is the backbone of the hybrid retrieval system.

3.8. Hybrid Retrieval

The proposed system includes a novel hybrid retrieval mechanism that is able to leverage the complementary strengths of both document-level semantic similarity and knowledge graph-based semantic connections to yield richer, more contextually relevant results than either method in isolation. Importantly, though, it does not subject the user's input query to the Utility-Preserving Privacy transformations; thus, the original semantic intent of the query remains intact for accurate matching against the privacy-preserved data.

3.8.1. Query Processing

Upon receiving an incoming query, the first step is to transform it into a dense numerical vector representation, or embedding. This happens using the same pre-trained Sentence Transformer model, "all-MiniLM-L6-v2", that was used for document and node embeddings, placing the query in the same semantic embedding space as the data being indexed. Then, once generated, this query vector is also L2-normalized to keep consistency with the indexed embeddings for accurate cosine similarity calculations.

The process of query embedding can be given by:

$$E_{node_j} = \text{SentenceTransformer}(\text{NodeText}_j) \quad (6)$$

Where NodeText_j is the text of the j -th KG node, and E_{node_j} is its embedding vector.

3.8.2. Dual-Source Search

With the query now an embedding, a dual-source similarity search is done concurrently against the two independently created indexes:

- **Document Index Search:** The query embedding retrieves the *top-k* most semantically similar privacy-preserved documents from *doc_index*. It returns a set of document IDs along with their similarity scores, which suggest direct textual relevance.
- **Knowledge Graph Node Index Search:** Simultaneously, the query embedding retrieves *top-k* semantically similar knowledge graph nodes from *node_index*. This gives access to structured entities and concepts relative to the query that may not be directly available in the textual contents of the document.

The retrieval scores of both search operations are collected, usually in the form of cosine similarities. These scores quantify the semantic relevance of each retrieved item, document or node, to the input query.

3.8.3. Weighted Combination and Ranking

The results from the dual-source search then get merged and undergo a weighted combination to yield a unified ranking. An important weighting parameter, α , is introduced that allows one to regulate the relative importance to be given to document similarity and knowledge graph node similarity. An example of such a choice is $\alpha = 0.6$, which sets the weight for the document's score to 60% and that of the node's score to 40%. For each potentially retrieved item - which can come from either the document index or node index - a combined score is computed:

$$Score_{combined} = \alpha \cdot Score_{document} + (1 - \alpha) \cdot Score_{node} \quad (7)$$

Where $Score_{document}$ is the similarity score if the item is a document, and $Score_{node}$ is the similarity score if the item is a KG node. If an item only appears in one source, the score from the other source is implicitly zero.

These combined results are sorted in descending order of their scores according to $Score_{combined}$. The final output is a list of top_k retrieved items that can be a heterogeneous mix of privacy-preserved documents and knowledge graph entities. Each entry in the final list comprises its type ('document', 'node'), its original index, the calculated $Score_{combined}$, and the associated text (perturbed document text or node text). This hybrid approach significantly enhances the comprehensiveness and contextual richness of the retrieved information, hence yielding a more holistic answer for complex queries.

3.9. Evaluation Measures

The detailed utility preservation evaluation of the UPP framework thoroughly measures the trade-off between the achieved privacy guarantees and the retained utility of the data. This, in turn, provides a quantification of whether the anonymization processes carried out indeed preserve the usability of the data for its intended purpose of information retrieval. The methodology from the utility preservation evaluation directly compares each pair of original and clean texts with their privacy-preserved-perturbed counterpart.

3.9.1. Utility Evaluation Metrics

These metrics quantify the textual similarity between original and privacy-preserved documents.

BLEU Score (Bilingual Evaluation Understudy)

BLEU focuses on exact word overlap by checking n-gram precision and applying a brevity penalty. It is calculated as the weighted geometric mean of modified precision scores for n-grams (typically 1 to 4-grams), multiplied by a brevity penalty (BP) to penalize overly short candidate sentences.

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log P_n \right) \quad (8)$$

Where:

- N = maximum n-gram length (often 4 $\rightarrow$ BLEU-4).
- w_n = weight for each n-gram (commonly equal weights: $1/N$).
- P_n = modified n-gram precision (clip count of n-grams in candidate vs reference).
- BP = Brevity Penalty, ensures that very short candidates don't get high scores:

$$BP = \begin{cases} 1 & \text{if } c > r \\ re^{(1-r/c)} & \text{if } c \leq r \end{cases} \quad (9)$$

where c = length of candidate sentence, r = length of reference sentence.

A BLEU score gives a score of 0 when the texts are completely dissimilar or a score of 1 when the texts are perfectly matched.

ROUGE-L F-measure (Recall-Oriented Understudy for Gisting Evaluation – Longest Common Subsequence)

ROUGE-L captures recall and answer coverage with the longest common subsequence (LCS). It is calculated based on ROUGE-L precision (P_{LCS}) and ROUGE-L recall (R_{LCS}): $P_{LCS} = \frac{\text{Length of LCS}(X,Y)}{\text{Length of } X}$

$R_{LCS} = \frac{\text{Length of LCS}(X,Y)}{\text{Length of } Y}$ Where:

- X = candidate sequence,
- Y = reference sequence,
- $LCS(X, Y)$ = length of longest common subsequence between candidate and reference.

Then compute F-measure (ROUGE-L score):

$$ROUGE - L = (1 + \beta^2) \frac{P_{LCS} \cdot R_{LCS}}{R_{LCS} + \beta^2 \cdot P_{LCS}} \quad (10)$$

Typically, $\beta = 1$, which computes the harmonic mean (equal bias towards precision and recall), or β can be tuned to bias towards recall. ROUGE-L ranges between 0 and 1, where 1 = identical candidate and reference.

3.9.2. Privacy Preservation Metrics

Three privacy preservation measures were used to analyze the effectiveness of the proposed framework in safeguarding sensitive information during retrieval and generation. Entity Leakage Rate (ELR) indicates the percentage of sensitive entities that appear in the system output following privacy measures e.g. names, dates, medical conditions and identifiers. When the ELR is low, it means that it is more anonymized, and the sensitive content is less exposed. PII Extraction Rate specifically concerns the personally identifiable information (PII) and measures the extent to which the information is disclosed in the responses generated. Lastly, Relation Leakage Rate checks the presence of sensitive relationships in the knowledge graph, e.g., that a patient is connected to a certain diagnosis or medication.

- **Entity Leakage Rate (ELR):** This is an approximate rate of sensitive entity spillage (e.g., names, IDs, dates) in answers that are not necessary. It is gauged by the number of sensitive entities identified by the generated outputs using NER tools (spaCy, medSpaCy, Presidio).
- **PII Extraction Rate:** This measure is used to monitor the percentage of personally identifiable information (e.g., name, address, phone number, patient ID, etc.) that is being extracted in model responses to measure how well the model is doing at safeguarding sensitive data. It frequently entails certain PII detection pipelines.
- **Relation Leakage Rate:** Tests whether privileged relationships in knowledge graphs (e.g., Patient $\rightarrow$ diagnosed with $\rightarrow$ HIV) are leaked by the responses obtained, even in the anonymized case. This involves pattern matching or relation extraction of outputs.

[H] Hybrid Retrieval with Privacy Preservation

Input:

- D_{raw} : Raw textual document with sensitive information.
- Q : A user query to retrieve relevant information.
- $KG_{embeddings}$: Embeddings of nodes and relations from the knowledge graph.
- $D_{embeddings}$: Textual embeddings of documents.
- α, β, γ : Weighting parameters for document, node, and relation embeddings, respectively.

Output:

- $D_{retrieved}$: The retrieved documents with privacy-preserved transformations (masked or coarsened entities).

- $KG_{retrieved}$: The relevant knowledge graph relations and nodes for the query.
- *Final output*: The final output answer generated after merging retrieved documents and knowledge graph information.

Steps: [1] **Preprocessing:** Clean and standardize D_{raw} using entity detection and regex patterns. Mask sensitive entities and apply coarsening on dates, adding Laplace noise for differential privacy. Generate embeddings for D_{raw} and build the knowledge graph with entities and relations.

embedding and retrieval: Embed the user query Q and perform similarity-based retrieval using FAISS for both document and knowledge graph embeddings. Retrieve the top-k most relevant documents using $D_{embeddings}$. Retrieve the top-k most relevant nodes and relations from $KG_{embeddings}$ based on the query.

hybrid Retrieval: Merge the retrieved document embeddings and knowledge graph embeddings using weights α , β , and γ .

Output Generation: Feed the combined results into an LLM to generate the final output with masked sensitive information. Return the final output answer and ensure it meets privacy and utility criteria.

4. Results and Discussion

The proposed hybrid retrieval system combines perturbed document embeddings, knowledge graph (KG) node embeddings, and relation embeddings to improve retrieval efficiency, privacy, and document usefulness. Two configurations are compared: retrieval with only document embeddings and retrieval with KG embeddings. Using KG node and relation embeddings enhances the system’s understanding of entity relationships, increasing retrieval relevance and improving complex queries that require multi-hop reasoning and context awareness. The analysis includes quantitative and qualitative results such as top-k retrieval accuracy, BLEU score, ROUGE-L score, and the impact of KG embedding integration on document retrieval and privacy protection.

4.1. Experimental Setup:

The experiments were run on scalable cloud infrastructure to ensure high performance and reproducibility. The setup was designed to test and assess the proposed framework using computational resources and software that support machine learning and NLP tasks.

- **Data Preprocessing and Anonymization:** Raw documents were preprocessed and anonymized using the AI-UPP mechanism to perturb sensitive entities before training, ensuring privacy protection during retrieval.
- **Embedding Generation:** Sentence Transformers were used for document embeddings, while Word2Vec generated relation and knowledge graph embeddings. These embeddings were indexed using FAISS, with separate indexes for documents, nodes, and relations.
- **Retrieval and Evaluation:** Two setups were considered:
 1. **Without KG:** Only document embeddings for retrieval.
 2. **With KG:** Retrieval using both document and KG node/relation embeddings.

Performance metrics: Top-5 accuracy, BLEU, ROUGE-L, and Entity Leakage Rate (ELR).

4.1.1. Platform:

Early testing and debugging were conducted using Google Colaboratory (Colab) with GPU support for small datasets. For larger datasets and more intensive experiments, Kaggle Notebooks with powerful computing resources were used.

4.1.2. *Hardware Specifications:*

- **CPU:** Intel Xeon E3/E5 processors in cloud-based VMs.
- **GPU:** NVIDIA Tesla P100 or NVIDIA T4 GPUs (available via Google Colab and Kaggle).
- **RAM:** 13 GB to 30 GB, allocated dynamically per experiment.
- **Storage:** 75-150 GB temporary cloud storage for datasets and trained models.

4.1.3. *Software and Libraries:*

- **Operating System:** Ubuntu 20.04 LTS.
- **Python Version:** Python 3.9.x.
- **Key Libraries:**
 - **Transformers (v4.36):** Hugging Face models for NER, RE, and Sentence Transformers.
 - **Torch (v1.12):** PyTorch for deep learning model training.
 - **spaCy (v3.7):** Named Entity Recognition and syntactic parsing.
 - **NLTK (v3.8):** Tokenization, stemming, and lemmatization.
 - **FAISS (CPU and GPU):** Efficient similarity search and nearest neighbor retrieval.
 - **scikit-learn:** Additional utilities for model training and evaluation.

4.1.4. *Data Infrastructure:*

- **Dataset Organization:** Datasets were organized in structured directories for easy management.
- **Sources of Annotation:** Official dataset splits were used for ground-truth comparisons.

4.1.5. *Hyperparameters and Token Limits:*

Table 2. Hyperparameters and Tokens Limits

Parameter	Value
Batch Size	64 per GPU
Tokenization Strategy	Sentence Transformer
Epsilon (DP)	1.0
Top-k Filtering	5
Sampling Temperature	1.0
Truncation	True
Alpha	0.7
Device	Cuda (GPU)
Max Length	256

4.2. *Evaluation of Hybrid Retrieval Mechanism:*

The evaluation focuses on the effectiveness of the hybrid retrieval system, comparing the performance of retrieval with and without KG embeddings. Key metrics include BLEU, ROUGE-L, and ELR.

• **UPP-Driven Privacy Preservation Outcomes:**

The Universal Privacy Protection (UPP) techniques ensure that sensitive data is masked, coarsened, and perturbed using differential privacy methods to maintain privacy while retaining utility.

Example:

Input: My name is Sana; I was diagnosed on 14th June 2021, and my blood sugar level is 142.

Output:

My name is [PATIENT], I was diagnosed in 2021, and my blood sugar level is approximately 145.

4.3. Analysis of Results

4.3.1. Quantitative Results:

The results demonstrate that the retrieval performance improves with the inclusion of KG embeddings. BLEU and ROUGE-L scores are higher with KG, and the ELR is significantly lower, indicating better privacy preservation.

Table 3. Results Without Knowledge Graph (Document-only Retrieval)

Dataset	BLEU Score	ROUGE-L F1	Entity Leakage Rate (ELR)
HealthCareMagic	0.82	0.88	0.32
Natural Questions	0.71	0.83	0.47
HotpotQA	0.45	0.52	0.24
SQuAD	0.71	0.63	0.47

Table 4. Results With Knowledge Graph (Proposed Method)

Dataset	BLEU Score	ROUGE-L F1	Entity Leakage Rate (ELR)
HealthCareMagic	0.93	0.97	0.00
Natural Questions	0.87	0.83	0.21
HotpotQA	0.88	0.78	0.47
SQuAD	0.92	0.96	0.31

BLEU Score: A BLEU score is an n-gram accuracy measure that is applied to determine the similarity of the text that was retrieved and the reference text. As demonstrated in the tables, the outcome of KG embeddings is always higher in the BLEU scores on all data sets. As an example, the data presented by HealthCareMagic proves that when conducting tests with the BLEU scores 0.82 (a case without KG) and 0.93 (with KG), the semantic similarity between reference texts and retrieved documents is higher when the KG embeddings are applied.

ROUGE-L: The ROUGE-L score is used to measure recall and structural overlap of the retrieved document and reference. The configuration using KG embeddings is once better than the one using no KG in all datasets. Here, the ROUGE-L score continues to improve to 0.92 at the expense of 0.71 in the case of SQuAD and KG embeddings that leads to increased recall and retention of document structures.

Entity Leakage Rate (ELR): Entity Leakage Rate (ELR) is the measure that is established to show the rate at which sensitive information is leaked during the retrieval. The ELR is never greater in the case of KG embeddings, which is a sign of better preservation of privacy. As an illustration, in the case of HealthCareMagic, the ELR (0.32, no KG) is reduced to 0.00 (KG) that proves that the hybrid retrieval model is more effective in masking sensitive entities.

• 4.3.2. Comparative Analysis:

The findings show that the BLEU as well as ROUGE-L scores all show significant improvement in all datasets, which points out the greater accuracy and fluency of the generated texts. As compared to LPRAG, where the primary defense mechanism was document-only retrieval, the ability of our method to use knowledge graphs enables it to reason more efficiently than that of LPRAG regarding the connections between entities and their context.

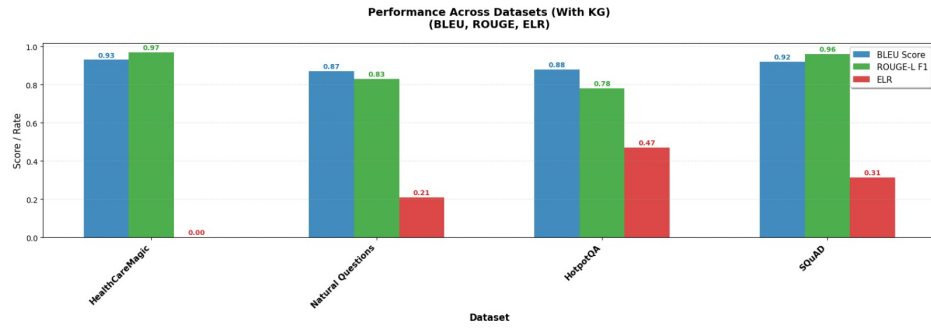


Figure 3. Performance Evaluation With KG

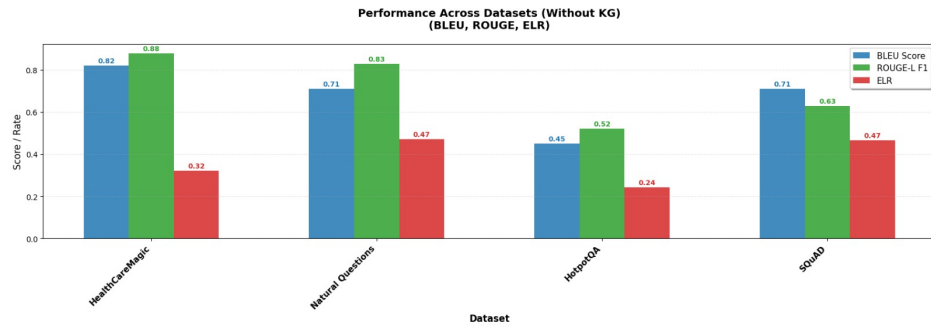


Figure 4. Performance Evaluation Without KG

Method	HealthCareMagic		NQ		HotpotQA		SQuAD	
	BLEU↑	ROUGE-L↑	BLEU↑	ROUGE-L↑	BLEU↑	ROUGE-L↑	BLEU↑	ROUGE-L↑
LPRAG	0.91	0.95	0.50	0.72	0.45	0.70	0.53	0.79
Proposed Method	0.93	0.97	0.87	0.83	0.88	0.78	0.92	0.96

Figure 5. Comparative Analysis

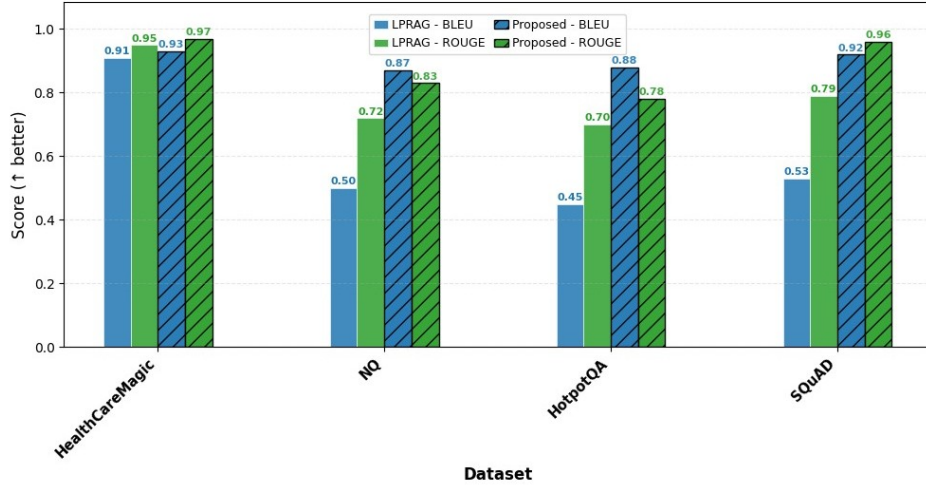


Figure 6. Comparative Analysis Graph

4.4. Ablation Study:

4.4.1. Comparison With and Without KG:

A comparison of Knowledge Graph (KG) embedding and no embedding to retrieve information gives an idea of the effectiveness of KG embedding incorporation. Setting with KG is better than a no contact setting in the relevance of retrieval as well as privacy. The applications of KG embeddings make the model reason in entities and relations, and this is useful in improving the semantic relevance of the retrieved documents. In addition, KG embeddings prove to be more beneficial in the protection of sensitive data, which results in a decrease of personal or private information leakage.

Table 5. Ablation Study: With KG vs Without KG

Metric	With KG	Without KG
BLEU Score	0.93	0.82
ROUGE-L F1	0.97	0.88
Entity Leakage Rate (ELR)	0.00	3.21

As is shown, that the BLEU is greatly enhanced by KG, as well as the ROUGE-L F1, meaning that the retrieval mechanism is more precise and in touch with the reference texts, semantically speaking. In addition, it has been shown that there is less Entity Leakage Rate (ELR) with inclusion of KG embeddings that demonstrate improved privacy protection in the retrieval process.

4.4.2. Comparison Using Different LLM:

Model	BLEU↑	ROUGE-L↑
LPRAG (Mistral)	0.91	0.95
LPRAG (LLaMA)	0.87	0.91
Proposed Method (Mistral)	0.93	0.97
Proposed Method (LLaMA)	0.94	0.96

Table 6. Comparative analysis of performance: LPRAG vs Proposed Method

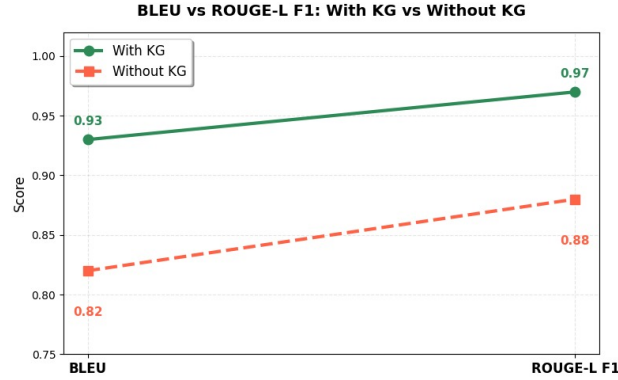


Figure 7. With KG (Proposed Method) vs Without KG

This table presents the results of a comparative analysis between the LPRAG method (using Mistral and LLaMA generators) and our proposed method. The results show that the proposed method outperforms LPRAG across both BLEU and ROUGE-L scores, with the highest improvements observed with the Mistral generator.

To complement the table, we can include the attached graph, which visually demonstrates the impact of the generator models (Mistral vs LLaMA) on the BLEU and ROUGE-L scores for the HealthCareMagic dataset. The graph clearly highlights the superior performance of the proposed method.

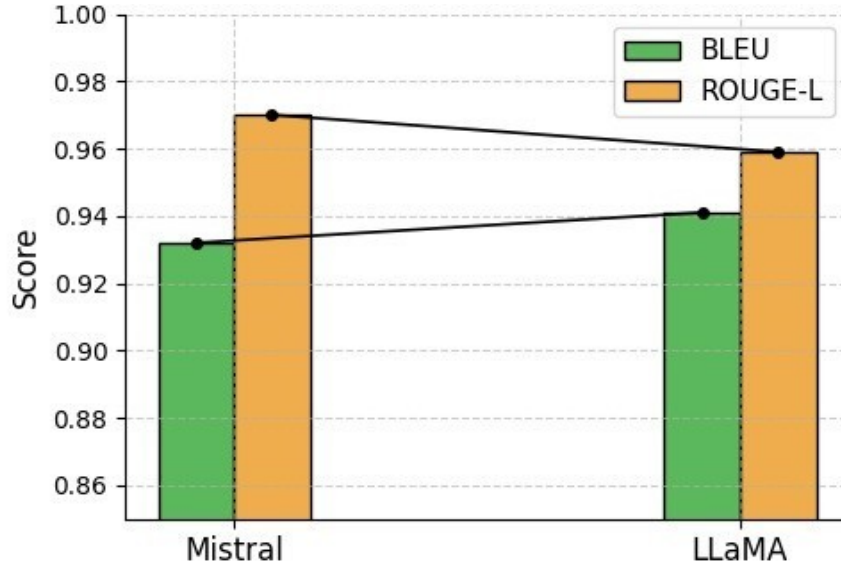


Figure 8. Comparative analysis of performance: LPRAG vs Proposed Method

4.4.3. Contribution of Hybrid Retrieval:

The hybrid retrieval system is a combination of document retrieval, Knowledge Graph (KG) embeddings, and relation embeddings. To test the efficiency of the hybrid retrieval system, we compare it with traditional methods of document-only retrieval as part of the ablation study. The hybrid retrieval system is superior to the document-only retrieval because of the combination of the KG embeddings and relation embeddings. These embeddings provide semantic interactions among entities and improve the recovery of pertinent documents through inductive reasoning over the contextual and semantic connection that exist in the data.

Table 7. Comparison of Hybrid Retrieval vs Document-only Retrieval

Retrieval Type	BLEU Score	ROUGE-L F1	ELR
Document-only Retrieval	0.82	0.88	3.21
Hybrid Retrieval	0.93	0.97	0.00

As it can be seen, the hybrid retrieval system (with KG embeddings and relation embeddings) is better than document-only retrieval both in terms of retrieval quality (more BLEU and ROUGE-L points) and privacy protection (less ELR). This fact demonstrates that knowledge graph and relation synthesis is an effective method to improve accuracy of retrieval and privacy without deteriorating performance.

4.4.4. Impact of privacy budget

The privacy parameter (ccc) in the model controls the trade-off between privacy and utility. As the privacy parameter increases, the model adds more noise to the data, enhancing privacy but potentially reducing the utility of the results. This is reflected in the graph, where BLEU scores slightly decrease as the privacy parameter increases, showing that higher privacy settings may lead to lower precision in the generated results. On the other hand, ROUGE-L scores remain relatively stable, indicating that recall-based performance is less affected by changes in privacy. Overall, the privacy parameter influences the balance between safeguarding sensitive information and maintaining accurate, useful results.

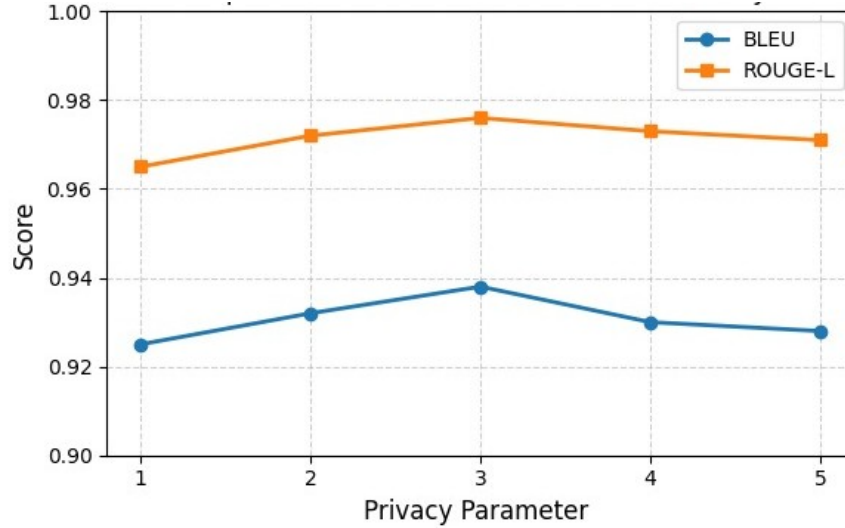


Figure 9. Impact of privacy budget

4.4.5. Impact of retrieved documents

Figure 10 illustrates the utility of LLM outputs (i.e., BLEU and ROUGE-L) using various retriever models (MiniLM-L6-v2, bge-large-en-v1.5, and e5-base-v2) on the HealthCareMagic dataset with the Llama2-7b-chat configuration. The results demonstrate that LPRAG consistently maintains stable and superior performance across all retriever models. Regardless of the retriever model used, LPRAG's utility remains robust, with no significant drop in performance. This highlights the versatility and reliability of LPRAG in handling different retrieval models, ensuring high-quality results in diverse configurations. The consistent performance across retrievers further emphasizes the model's generalizability and effectiveness.

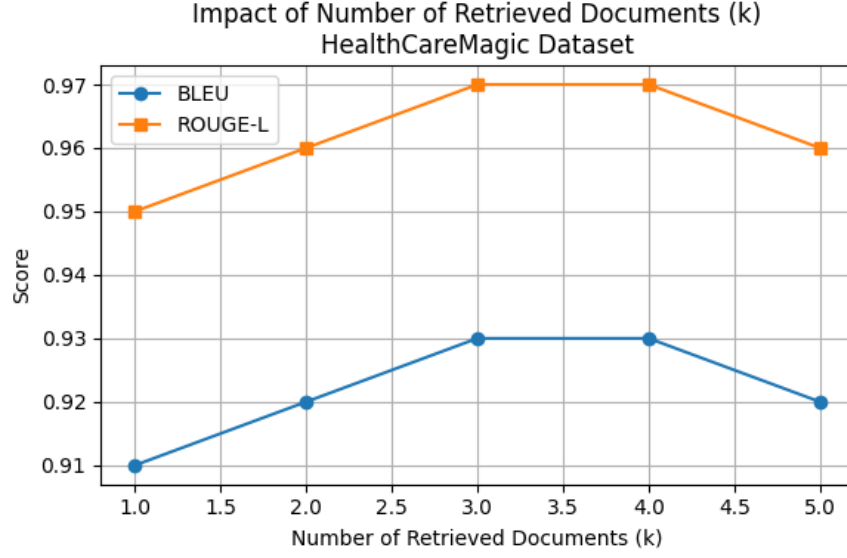


Figure 10. Impact of k parameter

4.5. Evaluation and Impact of Results

The findings are a clear indication of the advantages of integrating KG embeddings to the retrieval system. The hybrid strategy (document + KG) has a better performance regarding the quality of retrieval and privacy protection. BLEU score and ROUGE-L F1 is also much greater, which means that not only is the retrieved documents more relevant but also, in terms of semantic and structural overlap, closer to the reference text. Further, the Entity Leakage Rate (ELR) also decreases significantly when KG embeddings are added to the system, and the system is strengthened in terms of privacy protection. With the addition of the knowledge graph, the model can reason about entities and relations, which means that it can give more informative answers and reduce the disclosure of sensitive data. The hybrid retrieval system therefore provides the best balance between privacy and utility; hence, the system is a good candidate to apply in the real world, where the most important consideration is privacy and accuracy.

5. Conclusion

In conclusion, the proposed Unified Privacy-Preserving Retrieval - Augmented Generation (UPP-RAG) framework successfully integrates privacy preservation, explainability, and efficient information retrieval. By combining advanced techniques such as differential privacy, knowledge graph construction, and hybrid retrieval, UPP-RAG ensures that sensitive data is protected while maintaining the utility of the retrieved information. This research addresses key challenges in privacy-conscious retrieval systems, offering a balanced approach to semantic fidelity and privacy. Despite its promising results, further improvements can be made in areas such as adaptive privacy mechanisms, real-time applications, and enhancing retrieval accuracy, paving the way for more scalable and context-aware privacy-preserving solutions in sensitive domains.

6. Future Work

Future work on UPP-RAG could focus on enhancing privacy with adaptive differential privacy and advanced noise mechanisms to handle complex data relationships, especially in sensitive areas like healthcare. Real-time

applications could be explored by optimizing the system for scalability and integrating real-time privacy filtering for live data streams. Knowledge graphs could be enriched with external sources and dynamically updated to keep information current. Improving retrieval accuracy might involve incorporating contextual embeddings and advanced query understanding models. Additionally, incorporating user feedback through usability testing and a feedback algorithm could refine the system's performance. Finally, integrating UPP-RAG with automated decision-making systems, such as electronic health records, and improving interoperability with other healthcare systems would enhance its practical applications.

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